

Task: - practice the Data understanding, Data cleaning, EDA, Data preprocessing, Feature engineering and Feature selection

Process	What to Find / Analyze	California Housing (Regression)	Adult Income (Classification)	Wholesale Customers (Clustering)
Data Understanding	Dataset size	No. of rows & columns	No. of rows & columns	No. of rows & columns
	Feature types	Numerical vs categorical	Numerical vs categorical	Numerical vs categorical
	Target variable	MedHouseVal	Income	No target
	Business problem	What affects house prices?	What affects income level?	How customers can be segmented?
	Feature meaning	What each housing feature represents	What each demographic feature means	What each spending category means
Data Cleaning	Missing values	Which columns have nulls?	Which columns contain ? or nulls?	Any null values?
	Duplicates	Duplicate records?	Duplicate records?	Duplicate customers?
	Outliers	Extreme house prices?	Extreme age/hours?	Extreme spenders?
	Data types	Correct numeric types?	Correct categorical types?	Correct numeric types?
	Invalid values	Impossible room counts?	Invalid workclass/occupation?	Negative spending values?
EDA	Distribution	House prices distribution	Income distribution	Spending distribution
	Correlation	Which features affect price?	Which features relate to income?	Which spending categories move together?
	Patterns	Expensive vs cheap areas	High-income vs low-income people	Customer buying behavior
	Target analysis	Target distribution	Class balance	-
	Relationships	Feature vs price	Feature vs income	Feature vs feature
Data Preprocessing	Missing value strategy	Mean/Median?	Mode/Most Frequent?	Usually none
	Encoding	Rarely needed	One-Hot Encoding	Maybe Channel/Region
	Scaling	StandardScaler	StandardScaler	StandardScaler (very important)
Feature Engineering	Create new features	RoomsPerHousehold	AgeGroup	TotalSpend
	Ratio features	BedroomsPerRoom	HoursPerEducation	FreshRatio
	Aggregation features	PopulationPerHousehold	WorkIntensity	Spending Ratios
	Binning	HouseAgeGroup	AgeGroup	Customer Size Group
Feature Selection	Correlation analysis	Remove redundant features	Remove redundant features	Remove redundant features
	Mutual Information	Important predictors of price	Important predictors of income	Not usually used
	RFE	Best features for prediction	Best features for prediction	Not used
Task for after learning models and evaluation				
Model Training	Model comparison	Linear vs RF vs XGBoost	Logistic vs RF vs XGBoost	K-Means vs DBSCAN
	Hyperparameters	max_depth, n_estimators	max_depth, n_estimators	k, eps, min_samples
Model Evaluation	Key metrics	MAE, RMSE, R ²	Accuracy, Precision, Recall, F1	Silhouette Score
	Error analysis	Why predictions failed?	False Positives/Negatives	Badly clustered customers

	Feature importance	Most important predictors	Most important predictors	Important cluster drivers
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